Flyback Transformer Bobbin Fit – Full Explicit Calculations

This document verifies that all windings (split primary, secondary TIW, and auxiliary) fit on the RM10 bobbin when accounting for insulation tape between every layer, triple insulation at primary boundaries, and the split-primary extra tape requirement.

Given Parameters

Parameter	Value
Bobbin winding width	10.00 mm
Tape thickness	0.025 mm (25 µm polyester)
Primary wire	32 AWG magnet wire (OD = 0.24 mm)
Aux wire	34 AWG magnet wire (OD = 0.20 mm)
Secondary wire	TIW, 0.50 mm conductor
Secondary TIW OD	0.717 mm (worst case)
Primary turns	75T + 75T (split)
Secondary turns	6T
Aux turns	≤ 50T (single layer)

Width Fit Calculations

Winding	OD (mm)	Turns/Layer	Turns	Layers	Max Width Used (mm)
Primary (each half)	0.24	41	75	2	9.84
Secondary (TIW)	0.717	13	6	1	4.30
Aux (34 AWG)	0.20	50	≤50	1	≤10.0

Radial Build Calculations (Including All Tape)

Stack Element	Thickness (mm)
Primary half #1 (2 layers + tape)	0.505
Primary–Secondary boundary tape (triple + extra)	0.100
Secondary TIW layer	0.717
Tape between secondary and aux	0.025
Aux layer (34 AWG, 1 layer)	0.200
Tape between aux and primary	0.025

Secondary–Primary boundary tape (triple + extra)	0.100
Primary half #2 (2 layers + tape)	0.505
Total radial build	2.177 mm

Conclusion: All windings fit within the 10.00 mm bobbin width. Total radial build including all insulation and split-primary requirements is approximately 2.18 mm (with single-layer auxiliary winding).